Combinations

A **combination** is an arrangement of the items in a set in which the order of the items doesn't matter.

Suppose there are 8 workers in an office. The fire safety team needs 3 people. The manager puts the 8 workers' names on slips of paper and randomly chooses 3 names. Does the order the names are chosen make a difference?

No, the workers will be a group. The order doesn't matter.

Leticia		Joshua	
Calvin	=	Leticia	
Joshua		Calvin	

The orders are different, but the group of names is the same.

This is an example of a combination. Now let's see how to count it.

Example Hide

Three employees out of a group of 8 will be chosen randomly to be the fire safety team. How many different groupings are possible?

Step 1

Find the permutations for choosing 3 names out of 8.

$8 \times 7 \times 6 = 336$

Step 2

Find the permutations for a group of 3. You are finding how many different ways 3 names can be arranged.

$3 \times 2 \times 1 = 6$

Step 3

Divide.

$336 \div 6 = 56$

There are **56 possible combinations**.

How can you tell the difference between a permutation and a combination problem? You need to use common sense. Remember, if changing the order matters in the problem, it's a permutation.

Related Questions:
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